

Improper Fractions and Mixed Numbers

Sometimes a fraction will take a different form. A fraction with the top number equal to or greater than the bottom number is called an improper fraction.

When the numerator is *the same* as the denominator, the fraction *is equal to* 1.

Example 1

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$\frac{4}{4} = 1$ because 4 divided by 4 is 1.



When the numerator *is greater than* the denominator, the fraction *is greater than* 1.

Example 2

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$\frac{8}{5} > 1$ because 8 divided by 5 is 1 with a remainder of 3.



A **mixed number** is a “mix” of a whole number and a fraction. An improper fraction can be written as a mixed number using division.

Example 3

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Write $\frac{8}{5}$ as a mixed number.

Step 1

Divide the numerator by the denominator.

$\frac{8}{5} = 5 \overline{)8} \begin{matrix} 1 \text{ R} 3 \end{matrix}$

Step 2

The quotient becomes the whole number part.

Put the remainder over the original denominator to make the fraction part.

$\frac{8}{5} = 1 \frac{3}{5}$ *remainder*

Example 4

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Write $2\frac{1}{4}$ as a fraction.

Step 1

Multiply the whole number by the denominator. Write the product over the denominator of the fraction.

$2 \times 4 = 8$

$\frac{8}{4}$

Step 2

Add the original numerator to the product found in Step 1.

Write the total over the denominator.

$2\frac{1}{4} = \frac{8}{4} + \frac{1}{4} = \frac{9}{4}$

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